Connes' trace theorem on Carnot manifolds

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Introduction

In this talk I want to introduce my work in my pair of papers arXiv:2601.17794 and arXiv:2410.13701.
Mostly this gives me an excuse to introduce the whole theory of hypoelliptic operators, and so that is what I will do.

Part 1: Hypoellipticity (and some puzzles)

What is *hypo*ellipticity?

Let D be a differential operator on $\mathbb{R}^d$, say

$$D = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha.$$

We say that D is *hypoelliptic* if D has the property

$$Du \in C^\infty \implies u \in C^\infty.$$

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In particular:

- The kernel of D consists of smooth functions
- The eigenspaces of D consist of smooth functions

Reminder on *ellipticity*

A differential operator $D = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha$ is elliptic (of order m) if

$$\left| \sum_{|\alpha|=m} a_\alpha \xi^\alpha \right| \geq C|\xi|^m.$$

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Elliptic operators are hypoelliptic because of the "freezing principle": Write

$$\sum_{|\alpha| \leq m} a_\alpha(x)\partial^\alpha u(x) = f(x).$$

Where $f \in C^\infty$.

Reminder on *ellipticity* (cont.)

Pretend that the coefficients are constant, and invert D using the Fourier transform: we get

$$u_1(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} i^{-m} \Big(\sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha \Big)^{-1} \exp(i(x-y,\xi)) f(y)\, dy d\xi.$$

(some truncation of the singularity at $\xi = 0$ might be necessary.) It turns out that u_1 is smooth, and $u - u_1$ is smoother than u. This process can be iterated sufficiently many times to obtain any desired degree of smoothness on u.

Some puzzles

I will give some mysteries that have to do with hypoellipticity (although for the final two, the link is not obvious).

Puzzle 1: Kolmogorov's operator

Are there hypoelliptic operators that are not elliptic? Sure... On $\mathbb{R}^2 = \mathbb{R}_x \times \mathbb{R}_y$, we have the operator

$$K = -\partial_y^2 + y\partial_x.$$

Kolmogorov proved that K is hypoelliptic by the classic method of writing down a formula for K^{-1}.
Puzzle 1: What's going on here? Why is K really hypoelliptic?

Puzzle 2: Kohn's operator

Normally when we talk about elliptic operators $D = \sum_{|\alpha| \leq m} a_\alpha(x)\partial^\alpha$, the lower order part $\sum_{|\alpha| < m} a_\alpha(x)\partial^\alpha$ has no impact on the ellipticity. Hypoellipticity is very different...

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Hypoellipticity is very different...
Let $b \in \mathbb{C}$, and let $\mathcal{L}_b$ be the operator on $\mathbb{R}^3 = \mathbb{R}_x \times \mathbb{R}_y \times \mathbb{R}_z$ given by

$$\mathcal{L}_b = (\partial_x - \frac{1}{2}y\partial_z)^2 + (\partial_y + \frac{1}{2}x\partial_z)^2 + ib\partial_z.$$

Then:

- If $b \in \mathbb{C} \setminus (2\mathbb{Z}+1)$, then $\mathcal{L}_b$ is hypoelliptic
- If $b \in 2\mathbb{Z}+1$, then $\mathcal{L}_b$ is *not* hypoelliptic.

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Puzzle 2: Why does the first order term matter here? What causes this behaviour?

Puzzle 3: Lewy's operator

The Cauchy-Kovalevskaya theorem says that any PDE with analytic coefficients which is not obviously ill-posed is in fact locally solvable. For a long time, people expected something similar would be true for PDE with smooth coefficients. But it's not so.

Puzzle 3: Lewy's operator

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Let

$$L = \partial_x - i\partial_y + (ix + y)\partial_z, \text{ on } \mathbb{R}^3 = \mathbb{R}_x \times \mathbb{R}_y \times \mathbb{R}_z.$$

Lewy proved that for "most" C^∞ functions f, the equation $Lu = f$ is not locally solvable (meaning that for all x, there is no neighbourhood $U \ni x$ and $u \in C^\infty(U)$ such that $Lu = f$ on U).
Puzzle 3: What is so special about L? For which functions f is $Lu = f$ solvable?

Puzzle 4: Grushin's operator

Finally I will say something about Weyl's asymptotics. Let $D \geq 0$ be an elliptic operator of order m on a compact d-manifold M. Then D has discrete spectrum, consisting of eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots$. Weyl's asymptotics say that

$$N(\lambda, D) := \#\{\text{Eigenvalues of } D \leq \lambda\} \sim c\lambda^{\frac{d}{m}}, \quad \lambda \to \infty$$

for some constant c, which is determined by the principal symbol of D.

Puzzle 4: Grushin's operator (cont.)

Let

$$G = -\partial_x^2 - \sin^2(2\pi x)\partial_y^2, \text{ on } M = (\mathbb{R}/\mathbb{Z})^2 = (\mathbb{R}/\mathbb{Z})_x \times (\mathbb{R}/\mathbb{Z})_y.$$

This is not elliptic, but it is positive and it does have discrete spectrum. The eigenvalue counting function obeys

$$N(\lambda, G) \sim c\lambda \log \lambda, \quad \lambda \to \infty.$$

Puzzle 4: Where does the logarithm come from? And what is the meaning of the constant?

Part 2: Hörmander's sum of squares theorem

The Hörmander condition

Let $X_0, X_1, \ldots, X_n$ be smooth vector fields $\mathbb{R}^d$. $\mathrm{Lie}(\{X_1, \ldots, X_n\})$, the Lie algebra generated by $\{X_0, X_1, \ldots, X_n\}$ is the $C^\infty(\mathbb{R}^d)$-linear span of the iterated Lie brackets

$$[X_{j_1}, [X_{j_2}, \ldots, [X_{j_r}, X_{j_{r+1}}] \cdots]].$$

Definition

Say that $X_1, \ldots, X_n$ satisfy the Hörmander bracket condition $\mathrm{Lie}(\{X_0, \ldots, X_n\}) = \mathrm{Vect}(\mathbb{R}^d)$ (the set of all smooth vector fields on $\mathbb{R}^d$.) That is, every vector field is a C^∞-linear combination of iterated brackets of $\{X_0, \ldots, X_n\}$.

The Hörmander sum of squares theorem

Theorem

If $X_0, \ldots, X_n$ satisfy the Hörmander bracket condition, then the operator

$$X_1^2 + X_2^2 + \cdots + X_n^2 + X_0$$

is hypoelliptic.

The converse is "almost true", via the Frobenius theorem.

Example: Kolmogorov's operator

Let $X_1 = \partial_x$, $X_0 = x\partial_y$. Then

$$X_1^2 + X_0 = \partial_x^2 + x\partial_y.$$

We have $[X_1, X_0] = \partial_y$, so the span of $\{X_1, X_0, [X_1, X_0]\}$ contains every vector field.

Example: Grushin's operator

Let $X_1 = \partial_x$ and $X_2 = \sin(2\pi x)\partial_y$. We have

$$X_1^2 + X_2^2 = \partial_x^2 + \sin^2(2\pi x)\partial_y^2.$$

We have $[X_1, X_2] = 2\pi\cos(2\pi x)\partial_y$, and the functions $\sin(2\pi x)$ and $\cos(2\pi x)$ are linearly independent, so the span of $\{X_1, X_0, [X_1, X_0]\}$ contains every vector field.

Part 3: Analysis on graded Lie groups

Why graded Lie groups?

In the previous section, we spoke about Lie algebras generated by families of vector fields $X_1, \ldots, X_N$.
If we want to go deeper (and to consider operators of order higher than 2), then we need to build a pseudodifferential calculus. Graded Lie groups are a natural place to start.

Graded Lie algebras

A graded Lie algebra $\mathfrak{g}$ is a Lie algebra which admits a direct sum decomposition

$$\mathfrak{g} = \bigoplus_{n=1}^{\infty} \mathfrak{g}_n$$

where $[\mathfrak{g}_k, \mathfrak{g}_n] \subseteq \mathfrak{g}_{k+n}$.
The number

$$Q := \sum_{n=1}^{\infty} n\dim(\mathfrak{g}_n)$$

is called the homogeneous dimension of $\mathfrak{g}$.

Graded Lie groups

A graded Lie algebra is nilpotent, and so the Baker-Campbell-Hausdorff formula is terminating

$$\exp(X)\exp(Y) = \exp(X + Y + \frac{1}{2}[X, Y] + \cdots) = \exp(\mathrm{BCH}(X, Y))$$

where $\mathrm{BCH}(X, Y)$ is the Baker-Campbell-Hausdorff power series (actually a polynomial here).

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as the unique simply connected Lie group with Lie algebra $\mathfrak{g}$.
Actually, here exp is a homeomorphism, and G can be identified with $\mathfrak{g}$ with the group product

$$X \cdot Y = \mathrm{BCH}(X, Y).$$

Graded Lie groups (concretely)

In practice graded Lie groups appear as Lie algebras generated by vector fields with polynomial coefficients.
For example,

$$\mathfrak{h} = \mathrm{span}\{\partial_x - \frac{1}{2}y\partial_z, \partial_y + \frac{1}{2}x\partial_y, \partial_z\}$$

is the 3-dimensional Heisenberg Lie algebra. Another one is

$$\mathfrak{g} = \mathrm{span}\{\partial_x + \frac{1}{2}y^2\partial_z, \partial_y, y\partial_z, \partial_z\}.$$

Analysis on graded Lie groups

A graded Lie group G is a glorified Euclidean space. As a manifold, we have $G = \mathbb{R}^n$ and the Haar measure on G is nothing but the pushforward of the Lebesgue measure by the exponential map.

Analysis on graded Lie groups

A graded Lie group G is a glorified Euclidean space. As a manifold, we have $G = \mathbb{R}^n$ and the Haar measure on G is nothing but the pushforward of the Lebesgue measure by the exponential map.
Pseudodifferential operators on G are given by operators of the form

$$Tu(g) = \int_G K(g,h)u(gh)\,dh$$

for some singular kernel $K(g,h)$.

Left-invariant differential operators

The universal enveloping $\mathcal{U}(\mathfrak{g})$ consists of all polynomials in the generators $\{X_1, \ldots, X_m\}$ of $\mathfrak{g}$. Thinking of $X_1, \ldots, X_m$ as left-invariant vector fields, $\mathcal{U}(\mathfrak{g})$ consists of left-invariant differential operators.

Theorem (Helffer-Nourrigat)

Let $P \in \mathcal{U}(\mathfrak{g})$ be homogeneous (according to the grading $\mathfrak{g} = \bigoplus_{n=1}^{\infty} \mathfrak{g}_n$). Then P is hypoelliptic if and only if $\pi(P)$ is injective for all non-trivial unitary representations of G.

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The condition that $\pi(P)$ be injective for all $\pi \in \widehat{G} \setminus \{0\}$ is called a *Rockland* condition.

Example: The abelian case

Let $G = \mathbb{R}^d = \exp(\mathrm{span}\{\partial_1, \ldots, \partial_d\})$ be the abelian Lie group. The unitary representations are labelled by $\xi \in \mathbb{R}^d$, and

$$\pi_\xi(\partial_j) = i\xi_j.$$

A homogeneous element $P \in \mathcal{U}(\mathfrak{g})$ is just a constant coefficient differential operator

$$P = \sum_{|\alpha|=m} a_\alpha \partial^\alpha.$$

The Rockland condition is that

$$\pi_\xi(P) = i^m \sum_{|\alpha|=m} a_\alpha \xi^\alpha$$

is nonzero for all $\xi \neq 0$. This is just ellipticity.

Example: The Kohn Laplacian

On $\mathbb{R}^3$, consider the following vector fields:

$$X = \partial_x - \frac{1}{2}y\partial_z,\ Y = \partial_y + \frac{1}{2}x\partial_z,\ Z = \partial_z.$$

The Lie algebra $\mathfrak{h} = \mathrm{span}(\{X, Y, Z\})$ is a graded Lie group, with $\mathfrak{h} = \mathfrak{h}_1 \oplus \mathfrak{h}_2$, where $\mathfrak{h}_1 = \mathrm{span}(\{X, Y\})$ and $\mathfrak{h}_2 = \mathrm{span}(Z)$. This is the three-dimensional Heisenberg Lie algebra.
The Kohn Laplacian is

$$\mathcal{L}_b = X^2 + Y^2 + ibZ.$$

This is a left-invariant operator on $\mathbb{H} = \exp(\mathfrak{h})$, the three-dimensional Heisenberg group.

Example: The Kohn Laplacian (ctd.)

The group $\mathbb{H} = \exp(\mathfrak{h})$ is the three-dimensional Heisenberg group. Its irreducible unitary representions are

$$\widehat{\mathbb{H}} = \{\pi_{\alpha,\beta}\}_{\alpha,\beta\in\mathbb{R}} \sqcup \{\pi_s\}_{s\in\mathbb{R}^\times}.$$

Where $\pi_{\alpha,\beta}$ are the one-dimensional representations

$$\pi_{\alpha,\beta}(X) = i\alpha,\ \pi_{\alpha,\beta}(Y) = i\beta$$

and π_s are the infinite-dimensional Schrödinger representations

$$\pi_s(X) = |s|^{1/2}\partial_t,\ \pi_s(Y) = i\,\mathrm{sgn}(s)|s|^{1/2}t,\ \pi_s(Z) = is$$

on $L_2(\mathbb{R}_t)$.

Example: The Kohn Laplacian (ctd.)

We have

$$\pi_{\alpha,\beta}(\mathcal{L}_b) = -\alpha^2 - \beta^2$$

and

$$\pi_s(\mathcal{L}_b) = |s|(\partial_t^2 - t^2 - \mathrm{sgn}(s)b).$$

The Harmonic oscillator $\partial_t^2 - t^2$ has spectrum $\{1, 3, 5, 7, \ldots\}$ so we see that $\mathcal{L}_b$ is injective in every nontrivial irreducible representation if and only if $b \notin 2\mathbb{Z} + 1$.

Part 4: Analysis on Carnot manifolds and the tangent groupoid

Carnot manifolds

Definition

A *Carnot manifold* is a manifold M equipped with a filtration of sub-bundles of TM. That is, there are subbundles $(H^j)_{j=0}^N$ such that

$$0 = H^0 < H^1 < \cdots < H^N = TM$$

and if $E \in \Gamma H^j, F \in \Gamma H^j$, then $[E, F] \in \Gamma H^{j+k}$.

We should think of the directions in H^j as having "order j".

The osculating group

From the data of a Carnot manifold (M, H), we define the *associated graded bundle* $\mathfrak{t}_H M$, which is the graded vector bundle over M formed by

$$\mathfrak{t}_H M = \bigoplus_{n=1}^{N} (\mathfrak{t}_H M)^n, \qquad \text{where} \qquad (\mathfrak{t}_H M)^n = H^n / H^{n-1}.$$

If $X \in \Gamma(\mathfrak{t}_H M)^n$, and $Y \in \Gamma(\mathfrak{t}_H M)^m$, then $[X, Y]$ is a well-defined section of $(\mathfrak{t}_H M)^{n+m}$. Here, ΓE denotes the space of smooth sections of a vector bundle E. In fact more is true, for smooth functions f, g, we then have

$$[fX, gY] - fg[X, Y] \in \Gamma H^{n+m-1}.$$

This implies that the commutator of vector fields descends to a Lie bracket on the the fibres of $\mathfrak{t}_H M$. That is, $\mathfrak{t}_H M$ is a bundle of graded nilpotent Lie groups.

The fibrewise exponential $T_H M := \exp(\mathfrak{t}_H M)$ is called the *osculating group* of the filtration H.

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We should think of the $T_H M$ as being an infinitesimal model for M with its filtration H.

The homogeneous dimension

The number

$$Q := \sum_{k=1}^{\infty} k \cdot \mathrm{rank}(\mathrm{t}_H M^k) = \sum_{k=1}^{N} \mathrm{rank}(H^k)$$

is called the homogenous dimension of (M, H). This is the homogeneous dimension of the fibres of $T_H M$ as a graded Lie groups.

Illustrative example: Asteroids

Let $M = (\mathbb{R}/\mathbb{Z})^3 = (\mathbb{R}/\mathbb{Z})_x \times (\mathbb{R}/\mathbb{Z})_y \times (\mathbb{R}/\mathbb{Z})_\theta$, and consider the two vector fields

$$X = \cos(\theta)\partial_x + \sin(\theta)\partial_y, \quad Y = \partial_\theta.$$

The vector bundle $H^1 = \mathrm{span}\{X, Y\}$ gives M a depth 2 filtration,

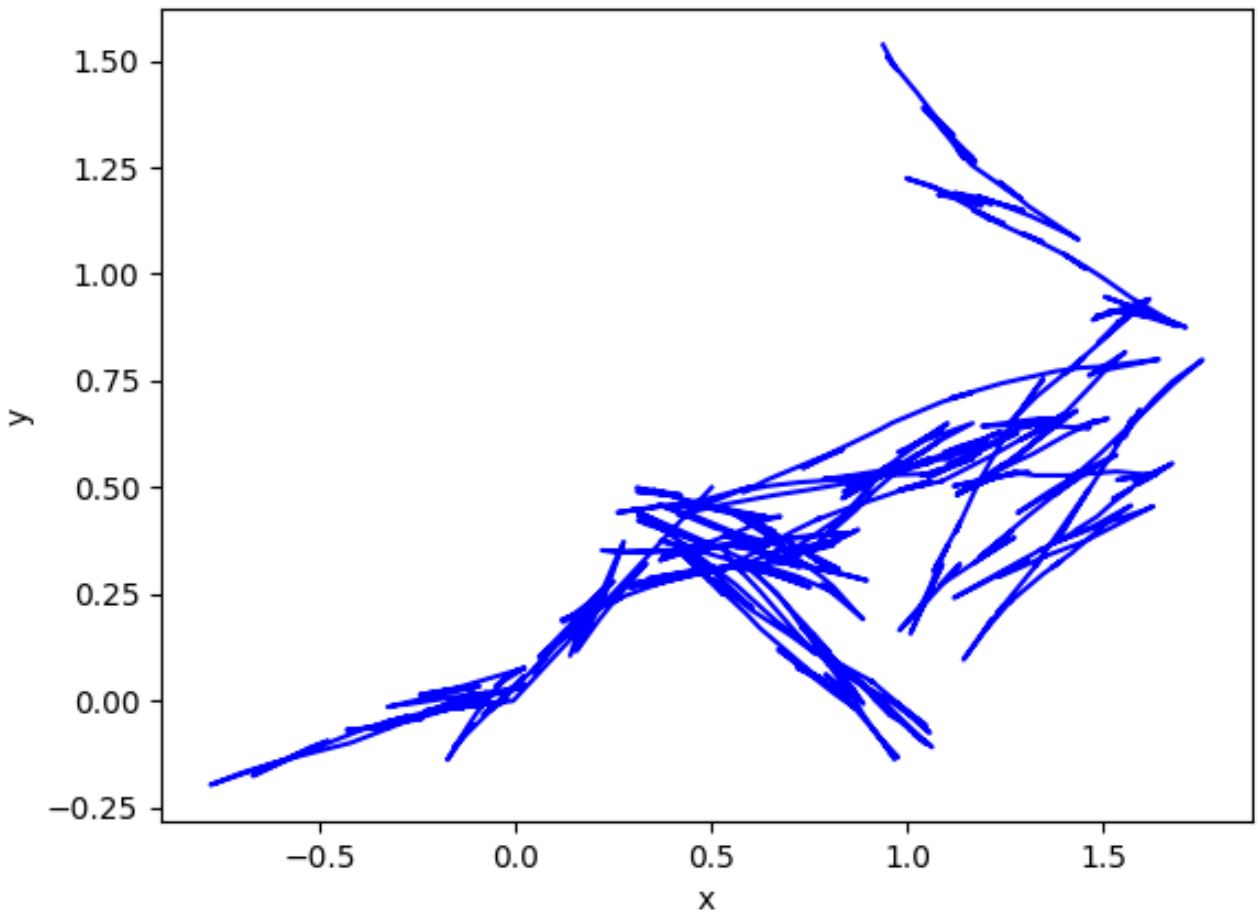
$$0 = H^0 < H^1 < H^2 = TM.$$

The osculating groups are all the three-dimensional Heisenberg group. The operator

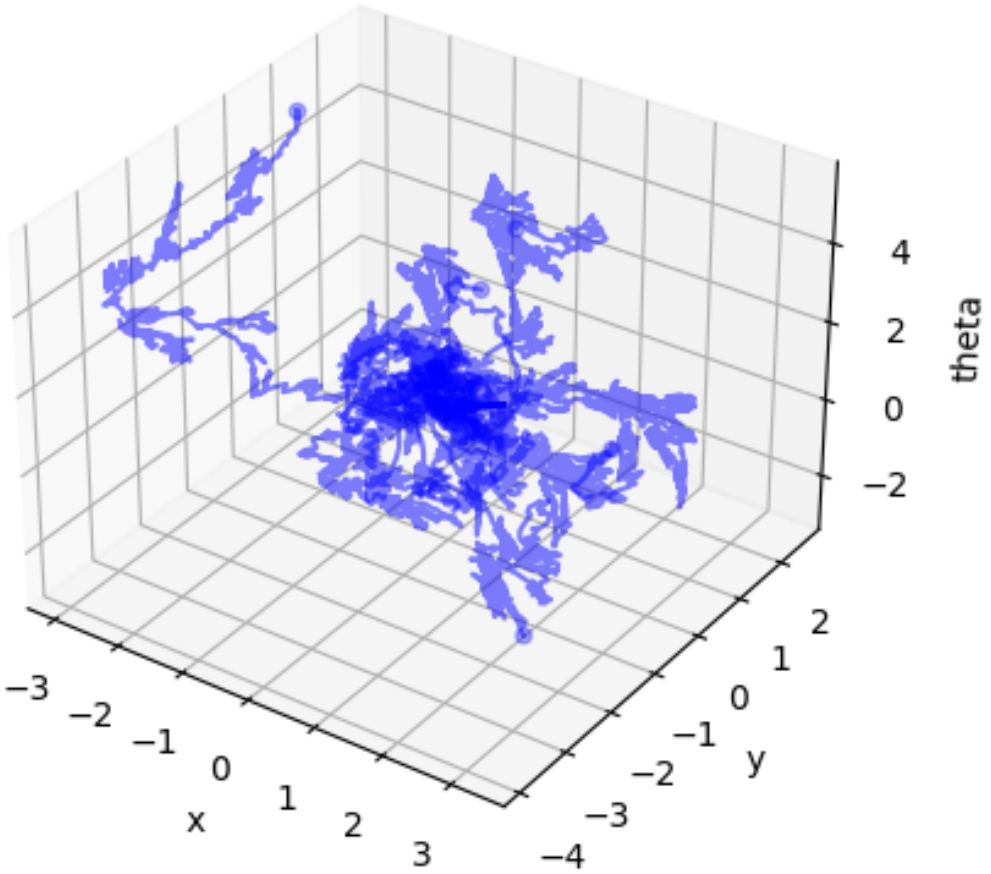
$$\Xi = X^2 + Y^2$$

satisfies the *pointwise Rockland property* and is therefore hypoelliptic.

A random path following X, Y

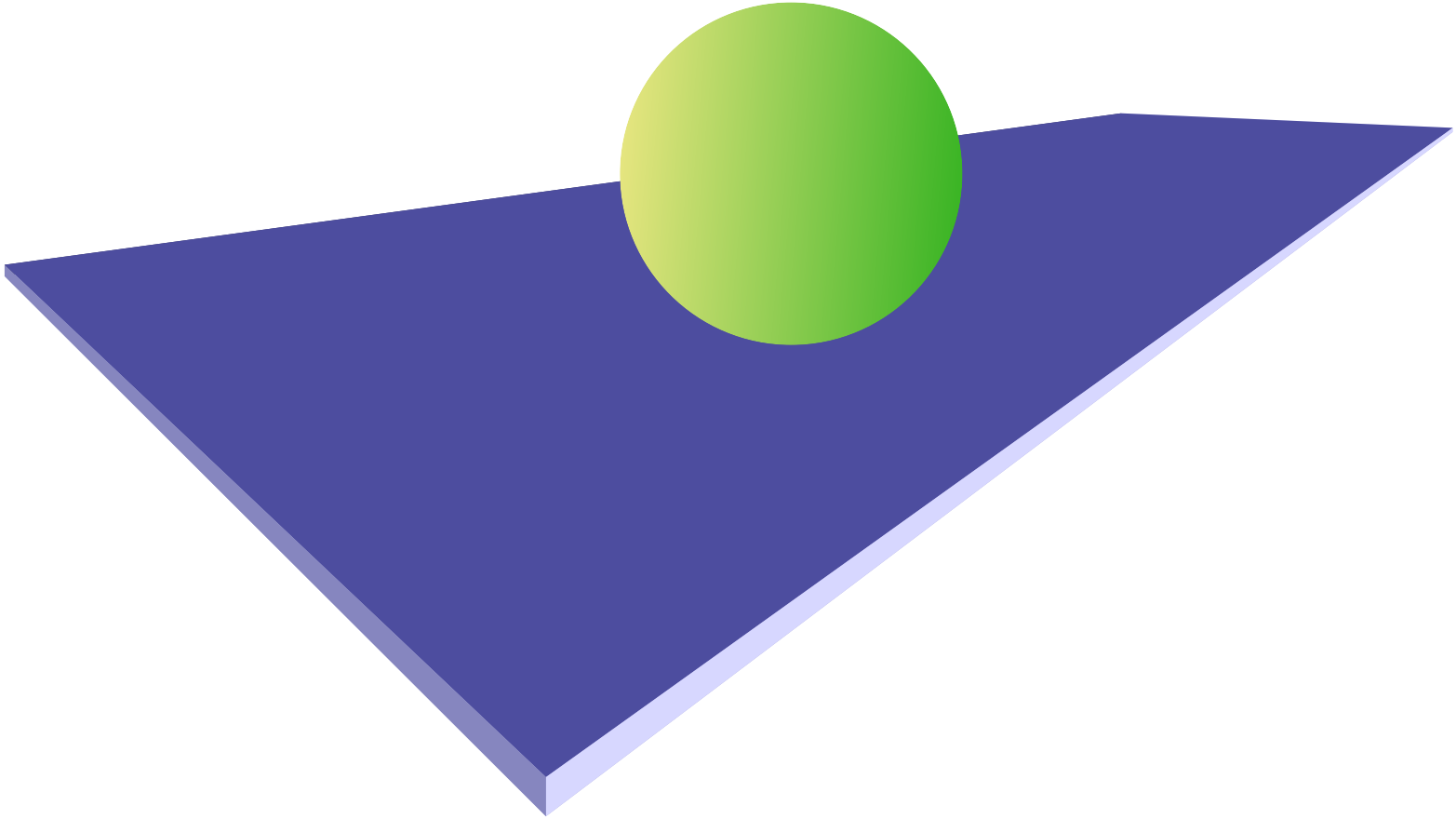


Plot of a collection of random paths following X, Y, in (x, y, θ) space



Another example: A rolling ball

Consider a ball rolling on a plane, without twisting or slipping:



Rolling ball example (cont.)

The space of orientations of the ball is $M = \mathrm{SO}(3)$. The Lie algebra of $\mathrm{SO}(3)$ is spanned by three vector fields:

$$X = \text{rolling around } x \text{ axis}, \ Y = \text{ rolling around } y \text{ axis}$$

and

$$Z = \text{rolling around } z \text{ axis.}$$

They satisfy $[X, Y] = Z$, $[Y, Z] = X$ and $[Z, X] = Y$.
To say that the ball "moves without twisting or slipping" is to say that the orientations of the ball move along the vector fields X, Y.

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To say that the ball "moves without twisting or slipping" is to say that the orientations of the ball move along the vector fields X, Y. Let

$$H^1 = \mathrm{span}\{X, Y\}.$$

Then

$$0 = H^1 < H^1 < H^2 = T\mathrm{SO}(3)$$

is a Carnot structure for $\mathrm{SO}(3)$, and $X^2 + Y^2$ is a hypoelliptic operator.

Pseudodifferential operators on a Carnot manifold

Pseudodifferential operators on a manifold M can normally be defined by

$$Tu(x) = \int_{TM_x} K(x,z)u(\exp_x(z))\,dz, \quad u \in C^\infty(M)$$

where K is the kernel of a pseudodifferential operator on tangent space TM_x. Something similar works for Carnot manifolds,

$$Tu(x) = \int_{T_H M_x} K(x,z)u(\Lambda_x(z))\,dz$$

where K is the kernel of a pseudodifferential operator on the graded Lie group $T_H M_x$, and Λ is a very carefully chosen family of local diffeomorphisms.

The right way to do it: The tangent groupoid

The main questions about defining pseudodifferential operators

$$Tu(x) = \int_{T_H M_x} K(x,z) u(\Lambda_x(z))\, dz$$

are:

i. What are the right conditions on the kernel function?
ii. What are the right conditions on the local diffeomorphisms Λ_x?

These are quite tricky problems... several solutions exist (they are more-or-less equivalent).

The right way to do it: The tangent groupoid

Van Erp and Yuncken proposed that the Schwartz kernels of pseudodifferential operators should be those which admit an approximately homogeneous extension to the H-tangent groupoid

$$\mathbb{T}_H M := T_H M \times \{0\} \sqcup M \times M \times \mathbb{R}^\times$$

(appropriately topologised).

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(appropriately topologised).
The advantages of their approach is that it is manifestly coordinate invariant, and it makes some of the elementary ψDO theory very easy.

Part 5: Conclusion and some results

Background on ideals

Let H be a Hilbert space, and let $\mathcal{K}(H)$ denote the compact operators on H. The trace ideal $\mathcal{L}_1$ is defined as

$$\mathcal{L}_1(H) := \{T \in \mathcal{K}(H) \ : \ \|T\|_1 := \sum_{n=0}^{\infty} \lambda(n, |T|)\}$$

and the weak-trace class ideal is defined as

$$\mathcal{L}_{1,\infty}(H) := \{T \in \mathcal{K}(H) \ : \ \|T\|_{1,\infty} := \sup_{n \geq 0}(n+1)\lambda(n, |T|)\}.$$

Background on traces

The trace ideal is so named because it's the natural domain of the operator trace

$$\mathrm{Tr} : \mathcal{L}_1(H) \to \mathbb{C}.$$

In general, a trace on an operator ideal $\mathcal{I}$ is a unitarily invariant functional

$$\varphi : \mathcal{I} \to \mathbb{C}, \quad \varphi(UAU^*) = \varphi(A), \qquad U \text{ unitary.}$$

Equivalently $\varphi(AB) = \varphi(BA)$ for all $A \in \mathcal{I}$ and bounded B.

The trace theorem

Theorem (Connes' trace theorem)

Let M be a compact d-dimensional manifold and let $T \in \Psi^{-d}(M)$ be a classical pseudodifferential operator of order $-d$. Let ν be a density on M. Then T can be identified with an element of $\mathcal{L}_{1,\infty}(L_2(M,\nu))$ and for any normalised trace φ on $\mathcal{L}_{1,\infty}(L_2(M,\nu))$ we have

$$\varphi(T) = \frac{1}{d}\mathrm{Res}(T).$$

The trace theorem (without traces)

Theorem (Connes' trace theorem)

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$$\sum_{n=0}^{N} \lambda(n, T) = \frac{1}{d}\mathrm{Res}(T)\log(N+2) + O(1), \quad N \to \infty..$$

Connes' trace theorem for Carnot manifolds

This is the obligatory part of the talk where I advertise my own results.

Theorem

Let (M, H) be a compact Carnot manifold with homogeneous dimension Q, and let $T \in \Psi_H^{-Q}(M)$ (in the van Erp–Yuncken sense). Then T can be realised as an operator belonging to the weak-trace-class ideal $\mathcal{L}_{1,\infty}$, and for any normalised trace φ on $\mathcal{L}_{1,\infty}$ we have

$$\varphi(T) = \frac{1}{Q}\mathrm{Res}(T)$$

where $\mathrm{Res}(T)$ is the residue as defined by Couchet–Yuncken.

Thank you for listening!